Week 3

Online Bias Learning (OBL)

with the Dow Data Challenge

Process Drift

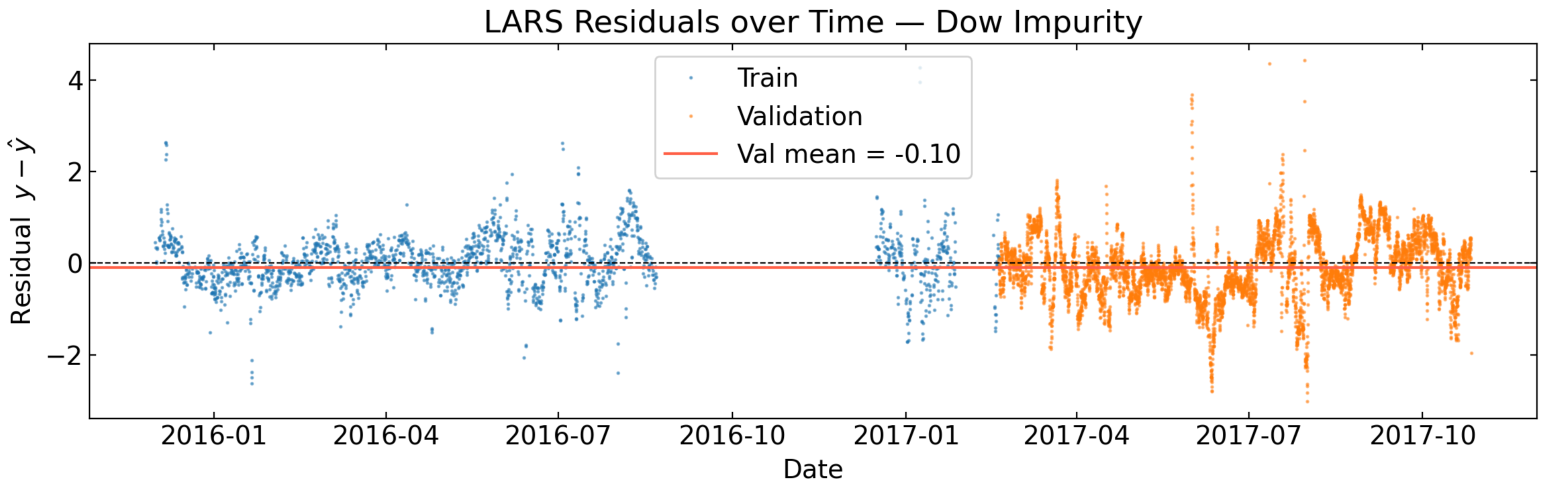
Statistical models are trained on historical data — but processes change over time:

- Equipment fouling, catalyst deactivation, feed composition shifts, seasonal effects

The model's predictions develop a **systematic bias** even if the model structure is correct.

Telltale sign: the mean residual on training data is zero by construction for OLS, but is **significantly nonzero** on validation data.

$$\frac{1}{n}\sum_t (y_t - \hat{y}_t) \approx 0 \quad \text{(train)} \qquad \frac{1}{m}\sum_t (y_t - \hat{y}_t) \neq 0 \quad \text{(validation)}$$



Online Bias Learning (OBL)

OBL (Qin Eq. 7–9) corrects the model prediction online as new lab measurements arrive.

Corrected prediction (Eq. 7):

$$\hat{y}_t = f(\mathbf{x}_t) + \hat{d}_t$$

$$\hat{d}_t = \frac{1}{n}\sum_{j=1}^{n}\left(y_{t-j} - f\left(\boldsymbol{x}_{t-j}\right)\right); \qquad \hat{d}_{t+1} = \frac{1}{n+1}\sum_{j=0}^{n}\left(y_{t-j} - f\left(\boldsymbol{x}_{t-j}\right)\right)$$

Using $\hat{y}_t = f(\mathbf{x}_t) + \hat{d}_t$, these two summations can be combined to give

$$\hat{d}_{t+1} = (1-K)\,\hat{d}_t + K(y_t - f(\mathbf{x}_t)); \qquad \hat{d}_{t+1} = \hat{d}_t + K\big(y_t - \hat{y}_t\big); \qquad K = \frac{1}{n+1}$$

The bias estimate is a weighted sum of **all** past raw residuals with exponentially decaying weights:

$$\hat{d}_t = K\sum_{i=0}^{t-1}(1-K)^{t-1-i}(y_i - f(\mathbf{x}_i))$$

OBL in Python

```python
def obl(model, X, y, K):
    if not 0 < K < 1:
        raise ValueError(f"K must be in (0, 1), got {K}")

    y_pred = np.zeros_like(y)
    d = np.zeros(len(y) + 1)       # d[t] is bias applied at step t

    for t in range(len(X)):
        y_hat = model.predict(X[t:t+1])[0]
        y_pred[t] = y_hat + d[t]                            # corrected prediction
        d[t+1] = (1 - K) * d[t] + K * (y[t] - y_hat)        # update bias

    return y_pred, d
```

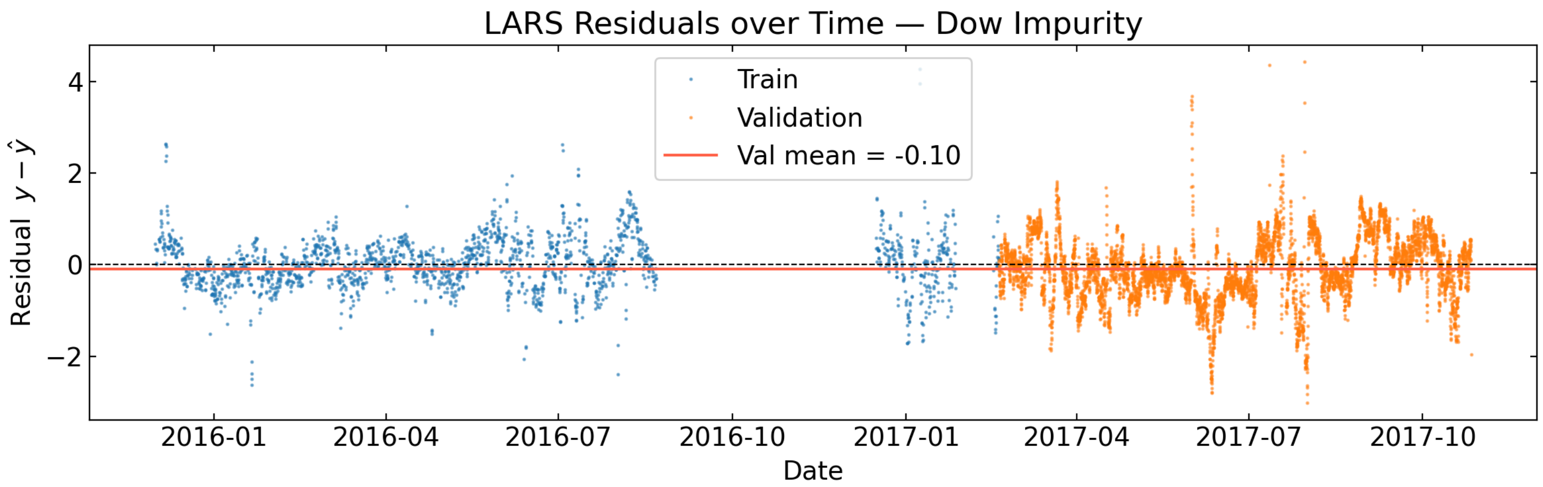
- `model.predict()` expects a 2D array, hence `X[t:t+1]`

OBL is a Model Correction

- **AFTER** we train the model, we apply the OBL to both training and validation steps.
- OBL is applied as a single forward pass from the start of training through validation
    - The bias estimate at the end of training carries over as the initial estimate for validation.

```
X = np.vstack([X_train, X_val])
y = np.concatenate([y_train, y_val])

y_pred_obl, d = obl(model, X, y, K)
```



Tuning K

Physical reasoning (Qin's approach)

- Drift occurs slowly, over many measurements. So average over long time scales.
- Qin: Use $n = 40$ most recent lab measurements → $K = 1/41 \approx 0.024$
- Restricted to $n \gg 1$ and $K \ll 1$
- Does not use validation data at all

Data-driven with split validation

- Tune K on the **first half** of validation
- Evaluate on the **second half**
- Avoids leakage (cannot use cross-validation — OBL is sequential)

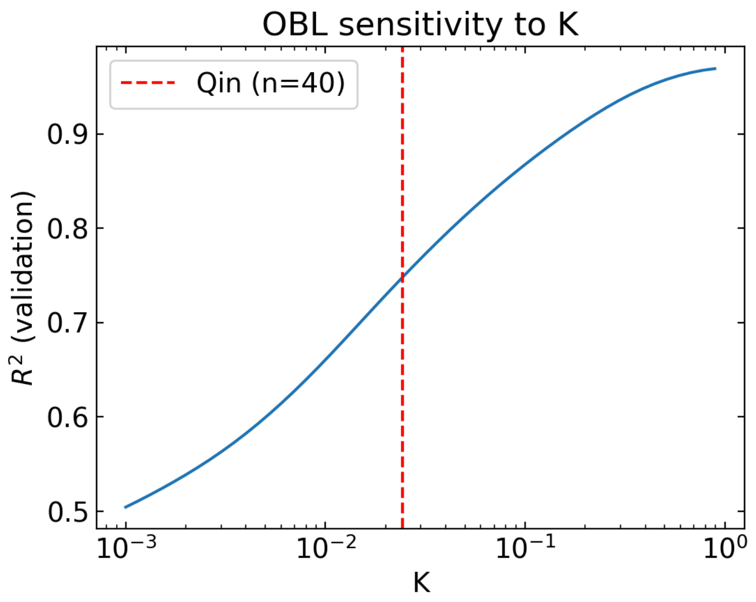
Warning: High K Degenerates to "Last Value"

As $K \to 1$: $\hat{d}_{t+1} \approx y_t - f(\mathbf{x}_t)$, so:

$$\hat{y}_{t+1} = f(\mathbf{x}_{t+1}) + y_t - f(\mathbf{x}_t) \approx y_t$$

for a slowly-varying process. **OBL with high K is just predicting the last measurement.**

Since impurity is autocorrelated, this gives artificially high R^2 — not model skill.



Naive baseline check

```
r2_naive = r2_score(y_val[1:], y_val[:-1])   # predict y[t-1] for y[t]
```

If OBL at the chosen K gives $R^2 \approx$ `r2_naive`, the model is doing no useful work.

Limitations of OBL

OBL corrects a **scalar bias** — it assumes the model structure is correct and only the intercept has drifted.

This breaks down when:

- The drift is **nonlinear** (e.g., sensitivity of impurity to a feed variable changes over time)
- The process moves into an **operating region not seen in training**
- The underlying physics changes (new reaction pathway, catalyst type)

A deeper question: can we build a model that is **less susceptible to drift** in the first place — by encoding physical knowledge directly into its structure?

Week 3

Physics-Informed Neural Networks (PINNs)

Embedding Physical Laws in Machine Learning Models

From Data-Driven to Physics-Informed

	Pure data-driven	OBL	PINN
Model structure	Learned from data	Fixed (LARS/PLS)	Constrained by physics
Drift correction	None	Bias update online	Physics reduces extrapolation error
Extrapolation	Poor	Poor (bias only)	Better — physics holds outside training range
Interpretability	Low–medium	Medium	High

Key idea: if the model respects physical laws (conservation of mass, energy, momentum), it is less likely to drift — physical constraints hold even when process conditions change.